Required Libraries: Install & Import

HDI Prediction System | Setup Reference

Overview

This document lists every third-party library used across the HDI Prediction System — the Flask web application, the model training notebook, and the deployment configuration — along with the exact install command and import statements needed to run each part of the project.

1. One-Step Install (Recommended)

All required libraries are listed in requirements.txt. From the project's Code folder, run:

```
pip install -r requirements.txt
```

Contents of requirements.txt:

```
Flask
numpy
pandas
scikit-learn
gunicorn
```

2. Library-by-Library Reference

Library	Install Command	Used In	Purpose
Flask	pip install Flask	app.py	Web framework: routes, request handling, rendering
numpy	pip install numpy	app.py, Training notebook	Building the feature array passed to the model; num
pandas	pip install pandas	Training notebook	Loading, merging, cleaning, and reshaping the HDI
scikit-learn	pip install scikit-learn	Training notebook (model training)	Training the regression model that predicts HDI fro
gunicorn	pip install gunicorn	Procfile / production deployment	Production-grade WSGI server used to run the Flas
pickle	Built into Python (no install needed)	app.py, Training notebook	Serializing/deserializing the trained model (HDI.pkl)
os, re	Built into Python (no install needed)	Training notebook	File system traversal and text pattern cleanup while

3. Imports Used in app.py (Flask Web App)

```
from flask import Flask, render_template, request, jsonify
import pickle
import numpy as np
```

Flask, render_template, request, jsonify handle routing, HTML rendering, reading form/JSON input, and returning JSON responses. **pickle** loads the trained model from HDI.pkl. **numpy** builds the 2D feature array expected by the model's predict() method.

4. Imports Used in the Training Notebook

```
import pandas as pd
import os
import re
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import pickle
```

pandas reads and merges the raw UNDP indicator files into HDI.csv and reshapes them into one row per country. **os** and **re** walk the input folder and clean file/column names. **numpy** supports numeric array operations during feature preparation. **scikit-learn** (train_test_split, LinearRegression or an equivalent regressor) trains the model that maps the four indicators to the HDI value. **pickle** serializes the final trained model to HDI.pkl for use by the Flask app.

5. Verifying the Installation

After installing, confirm everything imports correctly:

```
python -c "import flask, numpy, pandas, sklearn, gunicorn; print('All libraries imported successf
```

6. Recommended Environment

- Python 3.9 or higher.
- Use a virtual environment to avoid version conflicts: python -m venv venv && source venv/bin/activate (Linux/Mac) or venv\Scripts\activate (Windows).
- Run pip install -r requirements.txt inside the activated environment before starting app.py.